

offers the performance to handle complex software, making it a popular choice for designers. DSP cores such as Texas Instruments' Sitara family, or ARM family are both supported by extensive code libraries (such as OpenCV image processing library) for machine learning and image processing. These are all suitable for biometric recognition including faces and/or fingerprints. Debian is a Linux distribution that allows access to multiple programming languages including Python and C++, while also providing a useful environment for runtime execution.

For a long time, languages such as C and C++ have been the preferred choice for embedded systems, but things are changing and modern languages such as Python have valuable features that benefit designers. When working with multi-dimensional matrices and arrays, such as those necessary for machine learning, libraries with functions that support complex mathematical operations (such as Numpy library) are invaluable. In fact, using tools like these can dramatically simplify software development.

Operating with the ability to meet real-time deadlines is critical for all embedded applications, whether a simple door lock or a complex industrial motor controller. For this reason, MCU operating systems have always been optimised for software that guarantee this type of operational behaviour.

However, for all its popularity, most Linux distributions do not offer real-time operation, and those that do generally do not have sufficient access to the all-important libraries that software engineers desire. Multi-core implementations can help with this situation, as technologies such as Sitara family do not just offer DSPs that will run algorithms deterministically; these also encompass fast programmable real-time units that are capable of running separately to the

main ARM core that runs Linux.

Adding sophistication to the hardware can lower power consumption, especially if it is used to control the operation of power-hungry devices. In the door lock example, capturing images and processing them 24/7 is a waste of energy. A better approach would be to include a simple proximity sensor and only enable the camera and DSP when there is actually someone within the capture range.

An essential part of any development is experimenting and prototyping to see how different technologies and approaches can solve a problem. For example, several different technologies can be used for the door lock proximity sensor using pyroelectrics and/or ultrasonics. To do this, a simple development environment would be needed to allow peripherals to be interchanged quickly and easily.

One solution that has proven popular for prototype development is Beagleboard and Beaglebone platform. The physically-smaller Beaglebone is ideal for tight spaces, including for the door lock example. This versatile system allows the use of Click boards that make the attachment of different peripherals such as sensors and actuators extremely easy.

When designing simpler MCU-based devices there are several options available. However, when entering into the world of higher-performance embedded systems, options somewhat diminish. Even though these will

physically fit a Beaglebone board, the design process for powerful multi-core processors is much more difficult. If designers can source a host board with the requisite system-on-chip (SoC), then prototyping is easier, but the manufacturing optimisation phase must also be considered.

Often, a single power rail is all that is required for most MCUs, unlike SoCs and higher-performance microprocessors that frequently require several

different rails. Further, rails may need to be software-controlled to support low power modes but, once voltage is set, it must comply with tight tolerances—no matter how heavily it is loaded. Other complications faced by designers of these systems include the need to sequence power rails, often applying power to the core before other aspects of the SoC.

Many MCUs have onboard memory as their needs are relatively small. However, smart SoC-based systems need far greater resources, and designers need to consider the complexity that this adds to the project. This is especially true if the design involves image or video processing where the image or stream needs to be stored locally. When venturing into deep learning, requirements increase significantly to allow for storage of millions of parametric values as well as requirements for longer-term storage.

Provision for enough memory capacity is just one aspect of the challenge; speed needs to be sufficient, often necessitating the use of technologies such as DDR3 or DDR4. With fast memory, even the seemingly trivial task of routing the PCB can be difficult and lengthy. Tight timing requirements mean that trace lengths have to be equalised, even if that means somewhat tortuous routing snaking the shorter traces until these match the longer ones.

A convenient off-the-shelf solution to these design and layout challenges



Fig. 1: Functional block diagram for Octavo OSD335x SIPs